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Vellore Institute of Technology
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Course Code: CBS3005

**Course Title: Cloud, Microservices and
Applications**

Class Number: VL2025260105679

Lab Assessment - 1

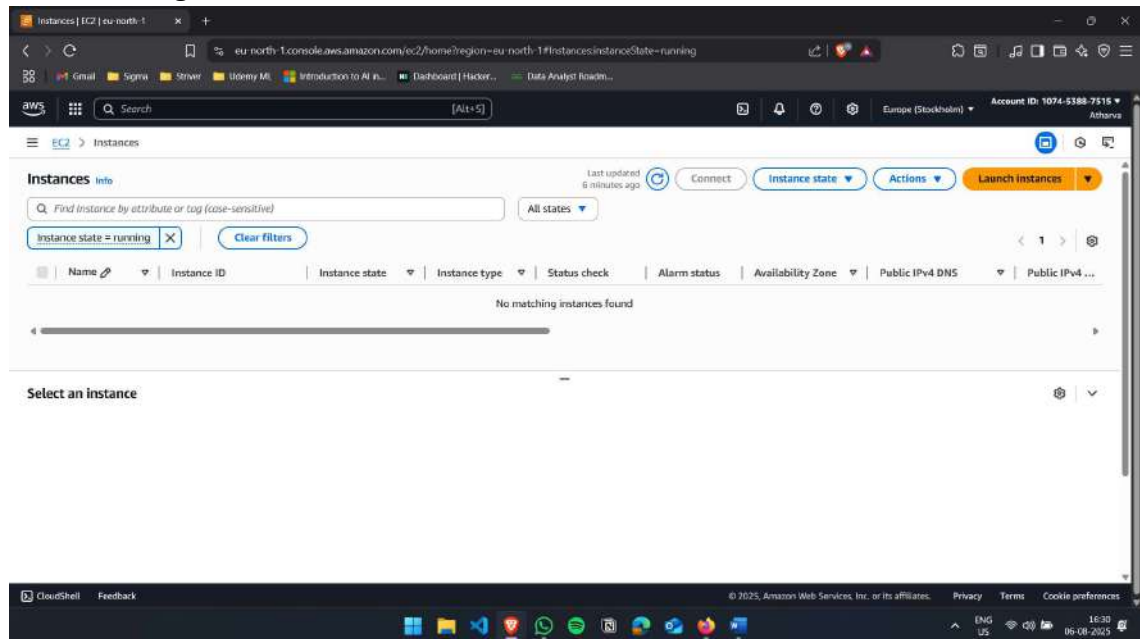
Name: Atharva Khachane

Reg No: 22BBS0155

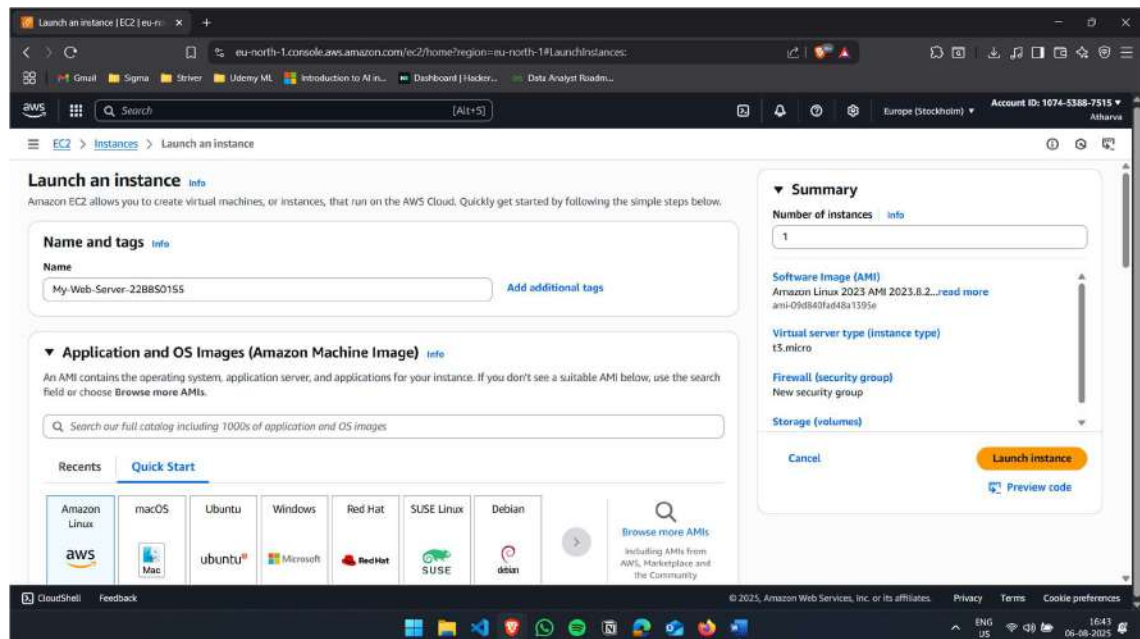
Q.1) Create and manage EC2 instances on AWS. Launch an application on an EC2 instance in one region and then migrate that instance to another region. Additionally, set up another EC2 instance to install MySQL, create a database (e.g., student or employee database), and perform basic SQL operations.

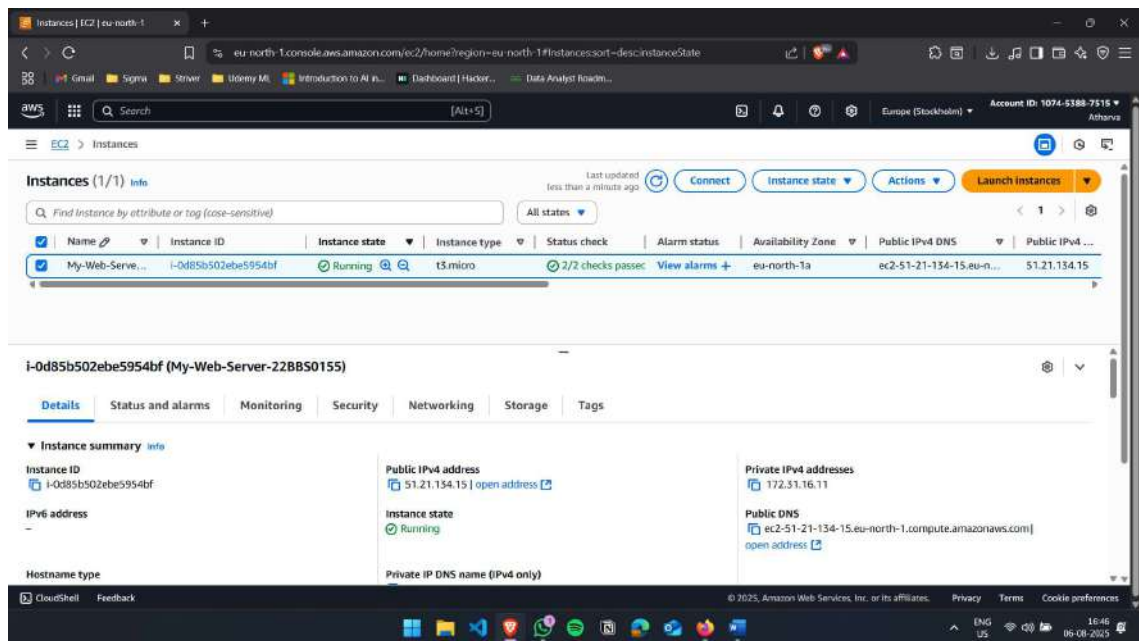
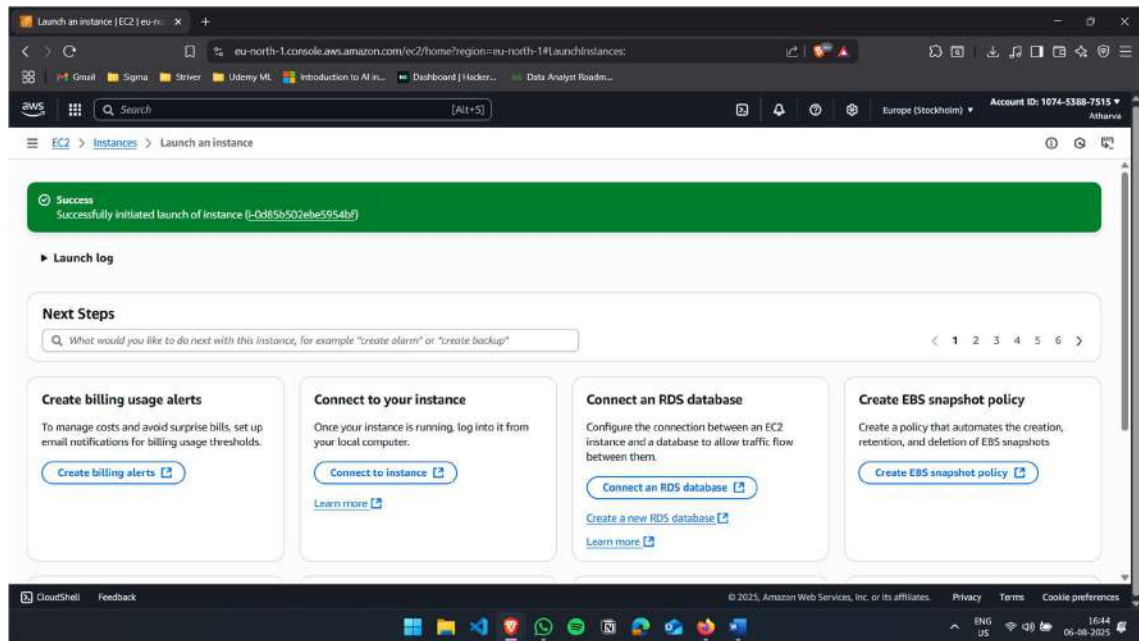
Solution:

- **Initial EC2 Page:**

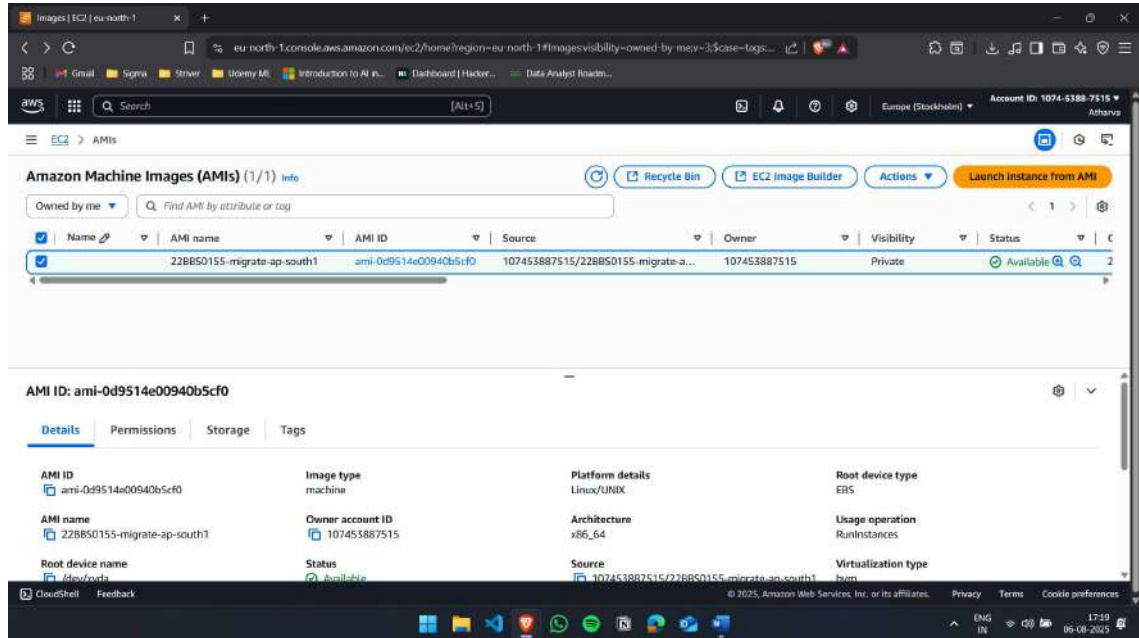


- **First EC2 Creation:**

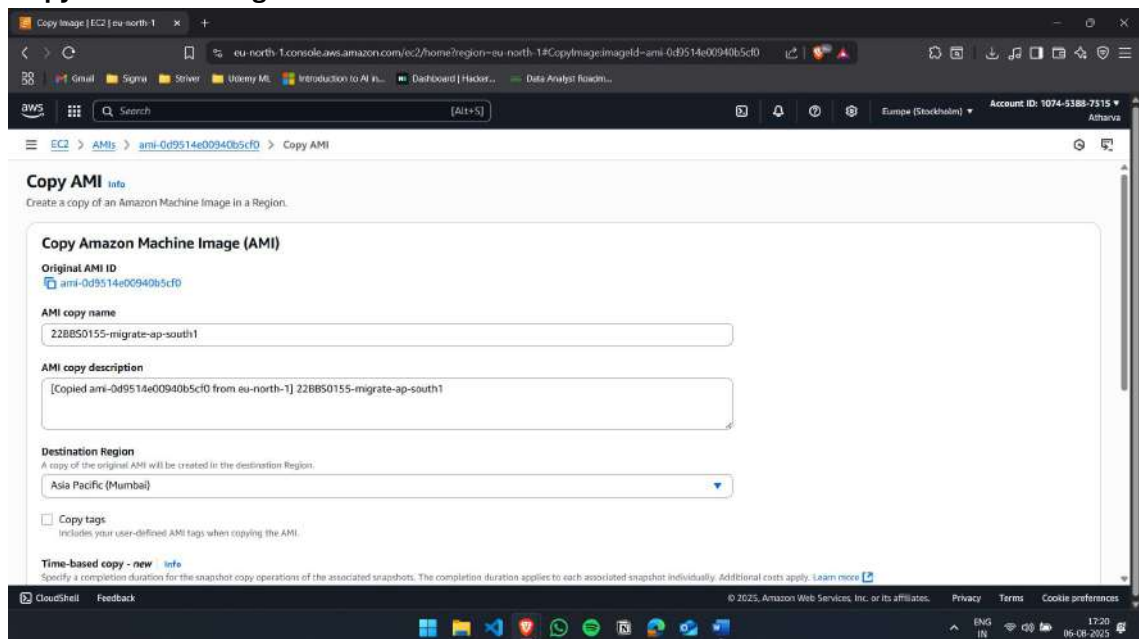




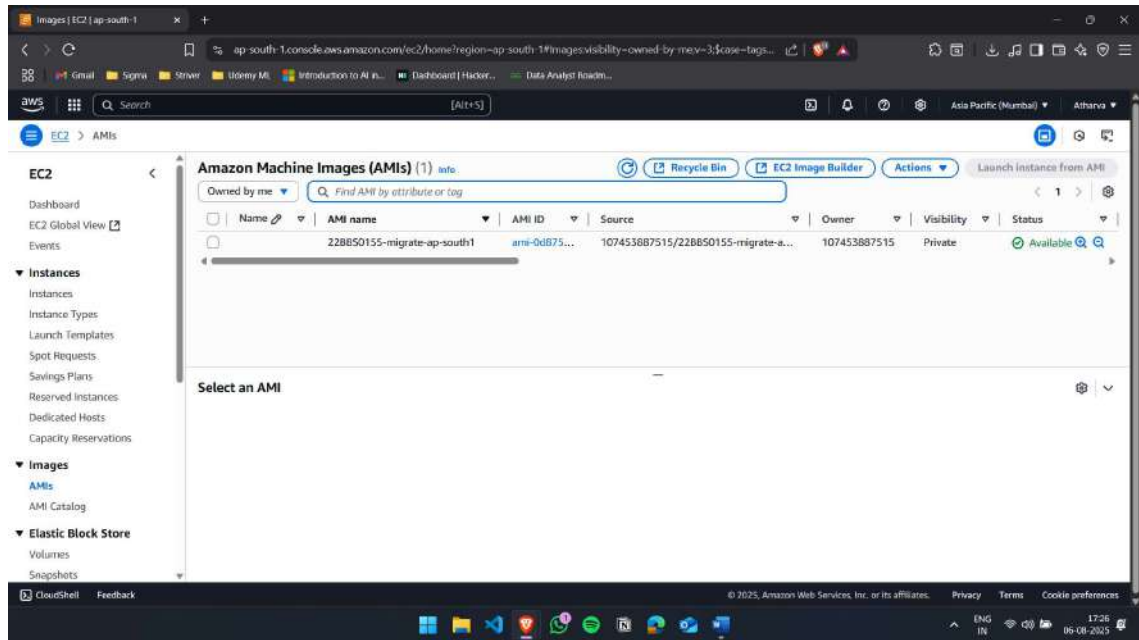
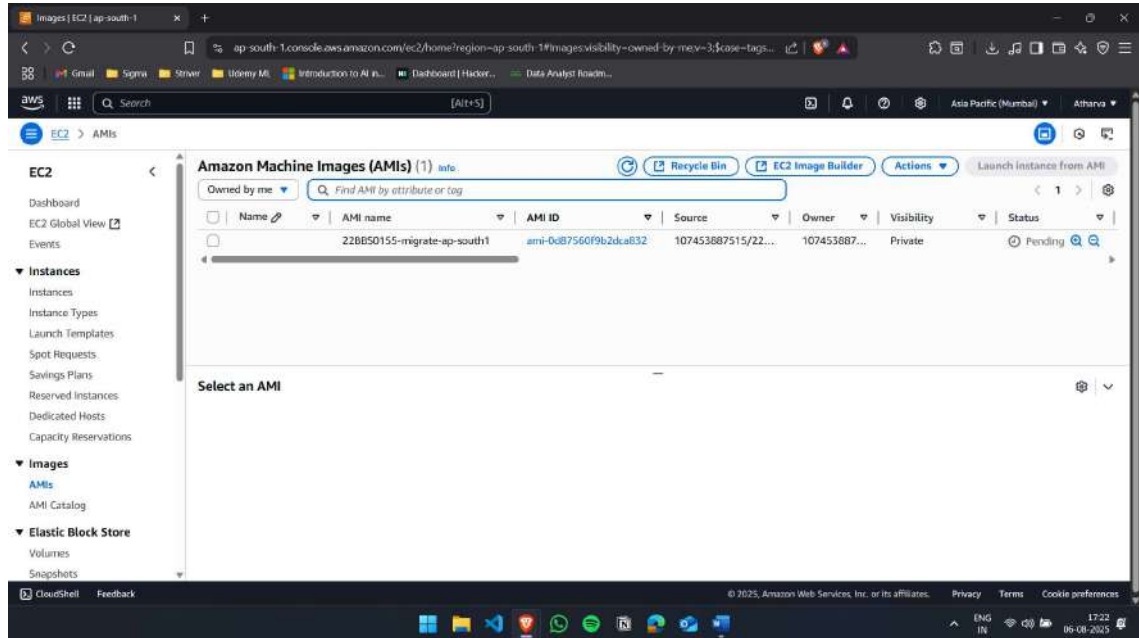
- **Create AMI of the Instance**



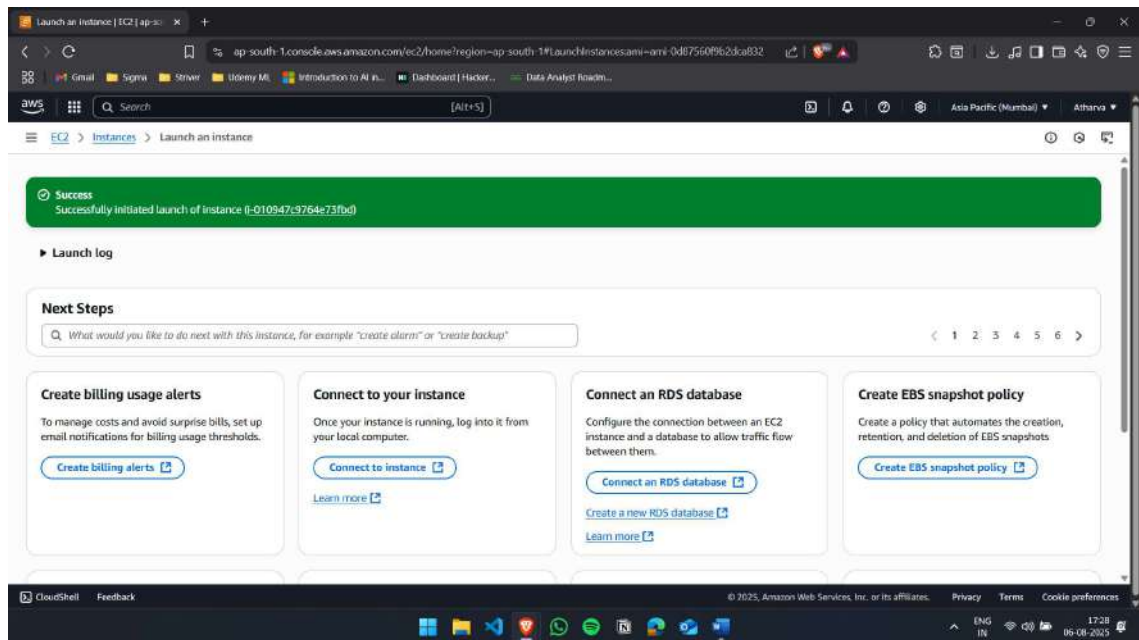
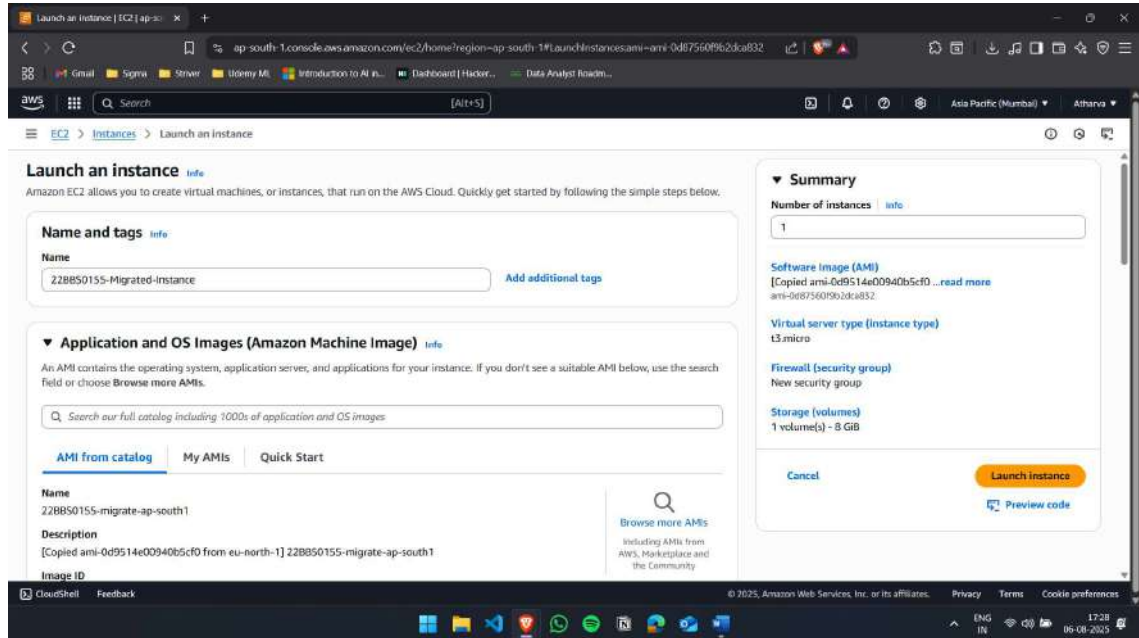
- **Copy AMI and Change the Destination to Mumbai**

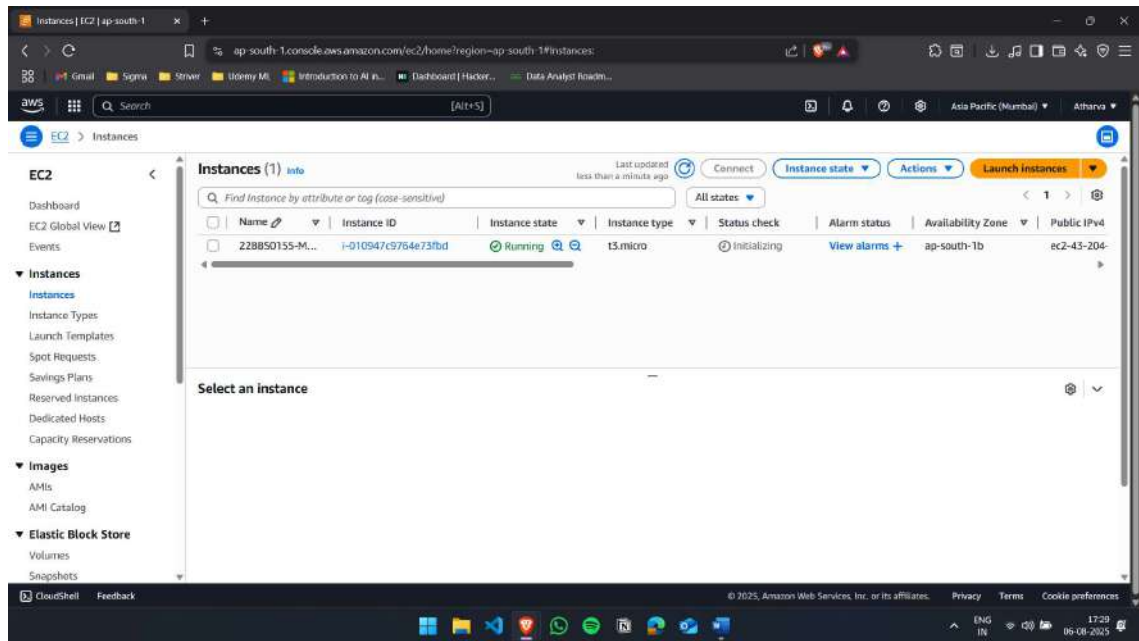


- Change the region to Mumbai

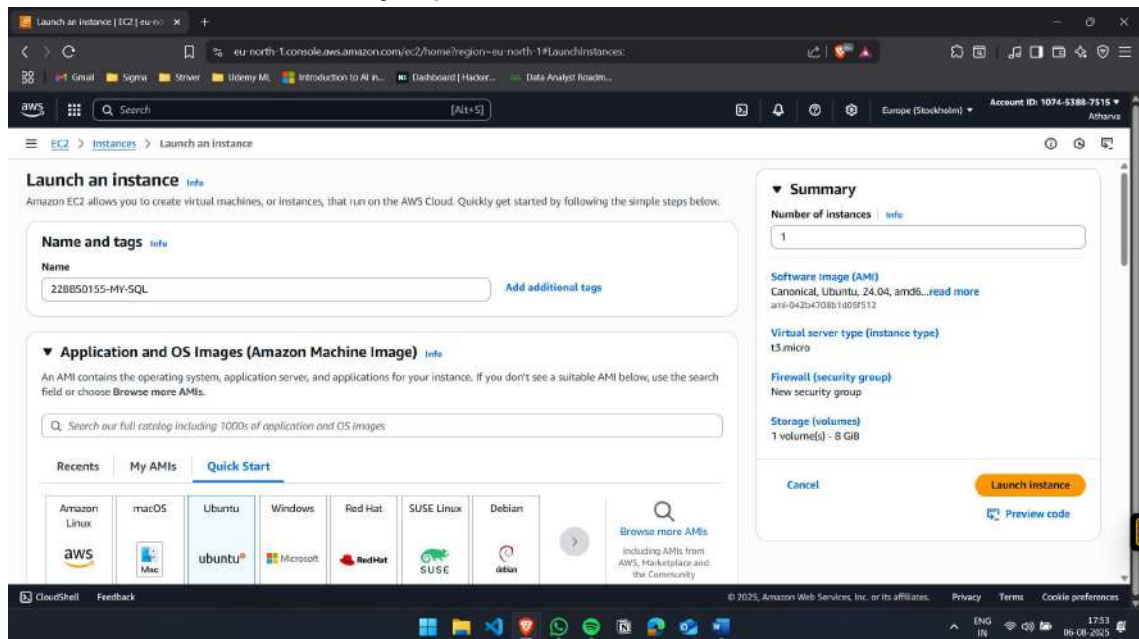


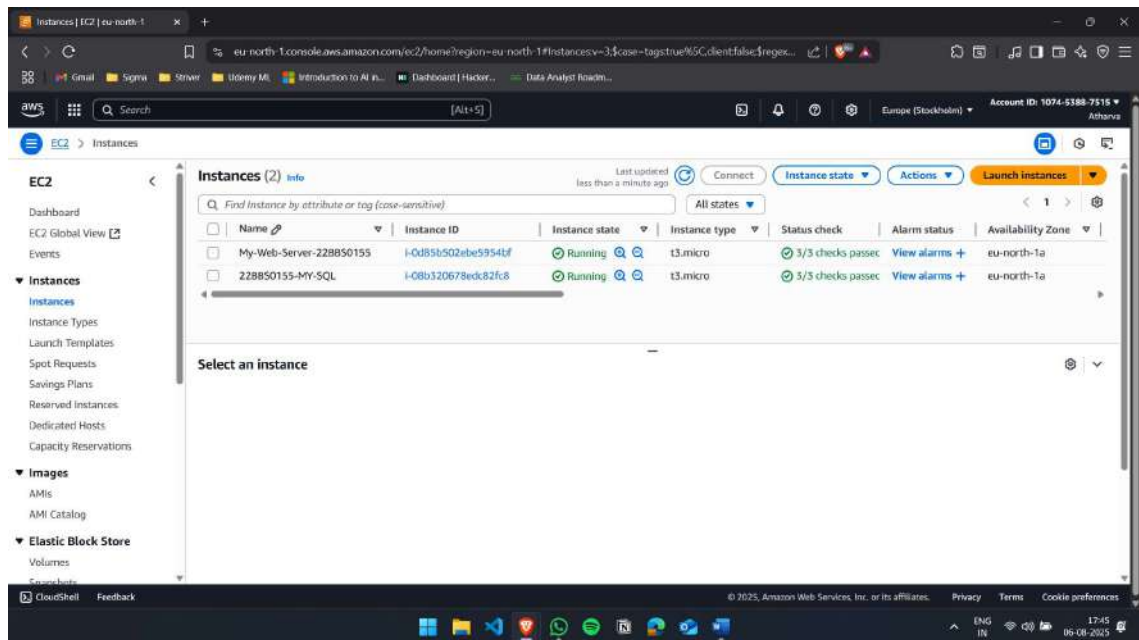
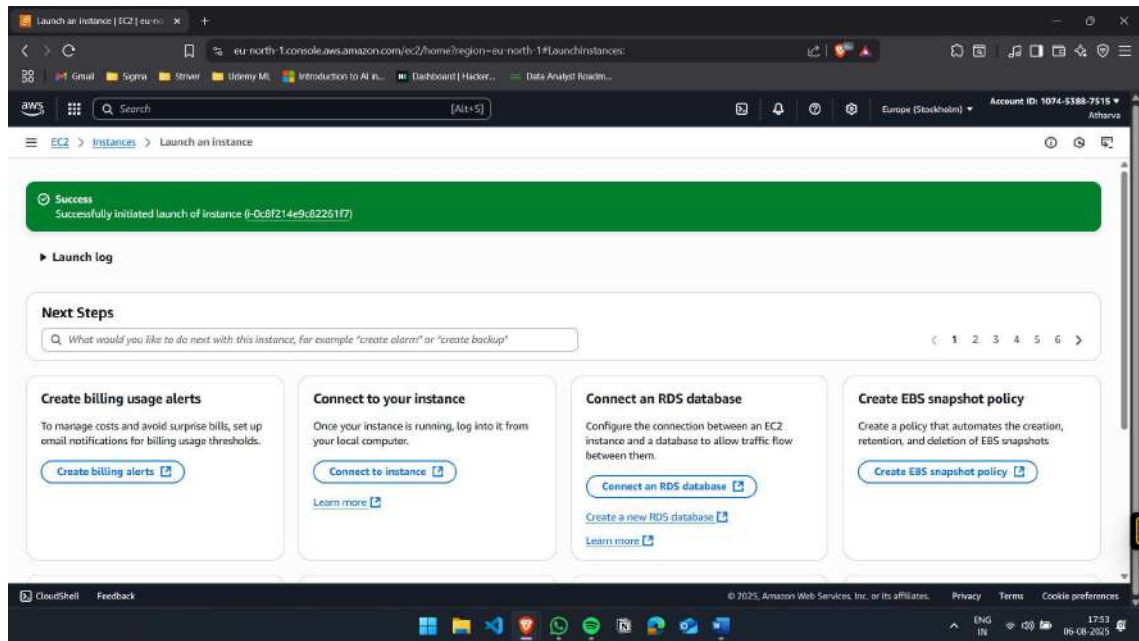
- Launch AMI Instance



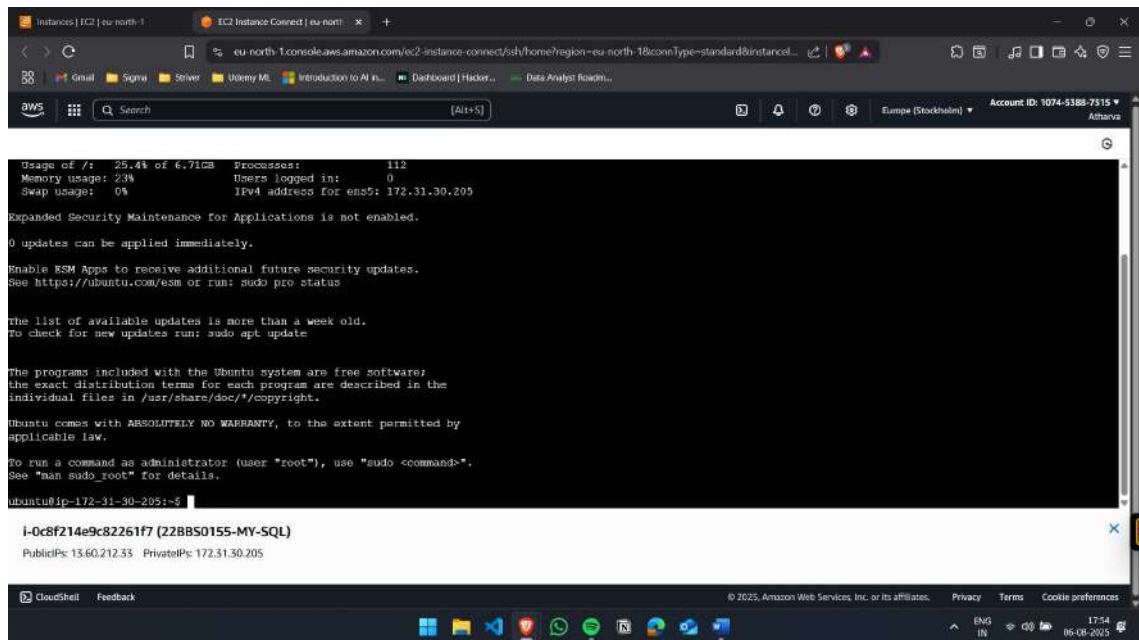


- Create another instance for My-SQL





- **Connect to the Instance**



The screenshot shows the AWS CloudShell interface with a terminal window. The terminal output displays system statistics: CPU usage at 25.4%, memory at 23%, and 112 processes. It also shows that no updates are available immediately and provides instructions to run 'sudo apt update' to check for updates. The terminal prompt is 'ubuntu@ip-172-31-30-205:~\$'.

```
Usage of /: 25.4% of 6.71GB    Processes:    112
Memory usage: 23%           Users logged in: 0
Swap usage: 0%              IPv4 address for ens5: 172.31.30.205

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

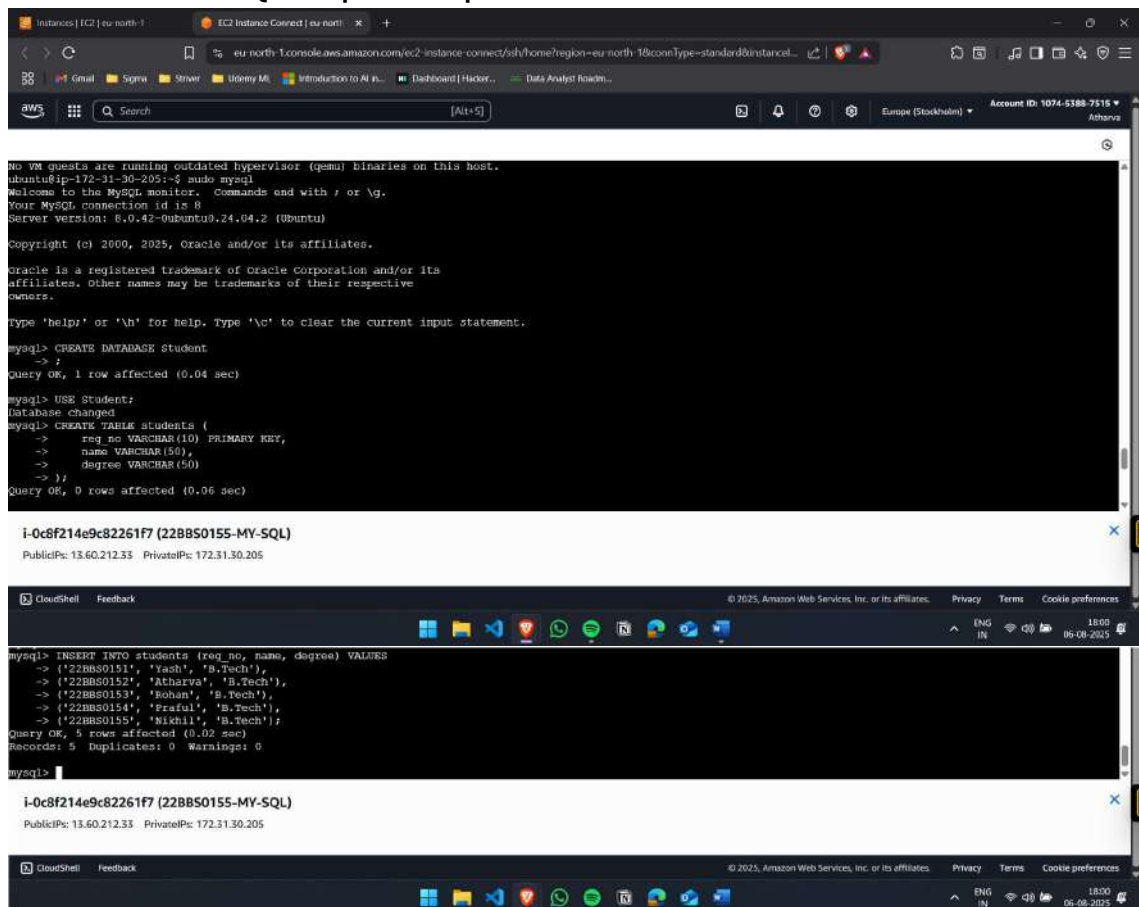
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-30-205:~$
```

Below the terminal window, the instance details are shown: **i-0c8f214e9c82261f7 (2288S0155-MY-SQL)**, Public IP: 13.60.212.33, Private IP: 172.31.30.205.

- **Install and run MYSQL and perform operations**



The screenshot shows the AWS CloudShell terminal with MySQL installed and running. The user runs 'sudo mysql' and is prompted with the MySQL welcome message. They then create a database named 'Student', use it, and create a table named 'students' with columns 'reg_no', 'name', and 'degree'. Finally, they insert five records into the 'students' table. The terminal prompt is 'mysql>'.

```
no VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-30-205:~$ sudo mysql
Welcome to the MySQL monitor.  Commands and with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.42-0ubuntu9.24.04.2 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE Student
->
Query OK, 1 row affected (0.04 sec)

mysql> USE Student;
Database changed

mysql> CREATE TABLE students (
->   reg_no VARCHAR(10) PRIMARY KEY,
->   name VARCHAR(50),
->   degree VARCHAR(50)
-> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO students (reg_no, name, degree) VALUES
->   ('2288S0151', 'Yash', 'B.Tech'),
->   ('2288S0152', 'Atharva', 'B.Tech'),
->   ('2288S0153', 'Rohan', 'B.Tech'),
->   ('2288S0154', 'Praful', 'B.Tech'),
->   ('2288S0155', 'Mishal', 'B.Tech');
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
```

Below the terminal window, the instance details are shown: **i-0c8f214e9c82261f7 (2288S0155-MY-SQL)**, Public IP: 13.60.212.33, Private IP: 172.31.30.205.

Instances | EC2 | eu-north-1

eu-north-1.console.aws.amazon.com/ec2-instance-connect/s3/home/region-eu-north-1?connectType=standard&instanceId=...

```
mysql> INSERT INTO students (reg_no, name, degree) VALUES
-> ('22BBS0151', 'Yash', 'B.Tech'),
-> ('22BBS0152', 'Atharva', 'B.Tech'),
-> ('22BBS0153', 'Rohan', 'B.Tech'),
-> ('22BBS0154', 'Praful', 'B.Tech'),
-> ('22BBS0155', 'Nikhil', 'B.Tech');
Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM students;
+-----+-----+-----+
| reg_no | name  | degree |
+-----+-----+-----+
| 22BBS0151 | Yash | B.Tech |
| 22BBS0152 | Atharva | B.Tech |
| 22BBS0153 | Rohan | B.Tech |
| 22BBS0154 | Praful | B.Tech |
| 22BBS0155 | Nikhil | B.Tech |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> DELETE FROM students WHERE name = 'Rohan';
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM students;
+-----+-----+-----+
| reg_no | name  | degree |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

i-0cbf214e9c82261f7 (22BBS0155-MY-SQL)

PublicIPs: 13.60.212.33 PrivateIPs: 172.31.30.205

CloudShell Feedback

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Instances | EC2 | eu-north-1

eu-north-1.console.aws.amazon.com/ec2-instance-connect/s3/home/region-eu-north-1?connectType=standard&instanceId=...

```
mysql> SELECT * FROM students;
+-----+-----+-----+
| reg_no | name  | degree |
+-----+-----+-----+
| 22BBS0151 | Yash | B.Tech |
| 22BBS0152 | Atharva | B.Tech |
| 22BBS0153 | Rohan | B.Tech |
| 22BBS0154 | Praful | B.Tech |
| 22BBS0155 | Nikhil | B.Tech |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> DELETE FROM students WHERE name = 'Rohan';
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM students;
+-----+-----+-----+
| reg_no | name  | degree |
+-----+-----+-----+
| 22BBS0151 | Yash | B.Tech |
| 22BBS0152 | Atharva | B.Tech |
| 22BBS0154 | Praful | B.Tech |
| 22BBS0155 | Nikhil | B.Tech |
+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

i-0cbf214e9c82261f7 (22BBS0155-MY-SQL)

PublicIPs: 13.60.212.33 PrivateIPs: 172.31.30.205

CloudShell Feedback

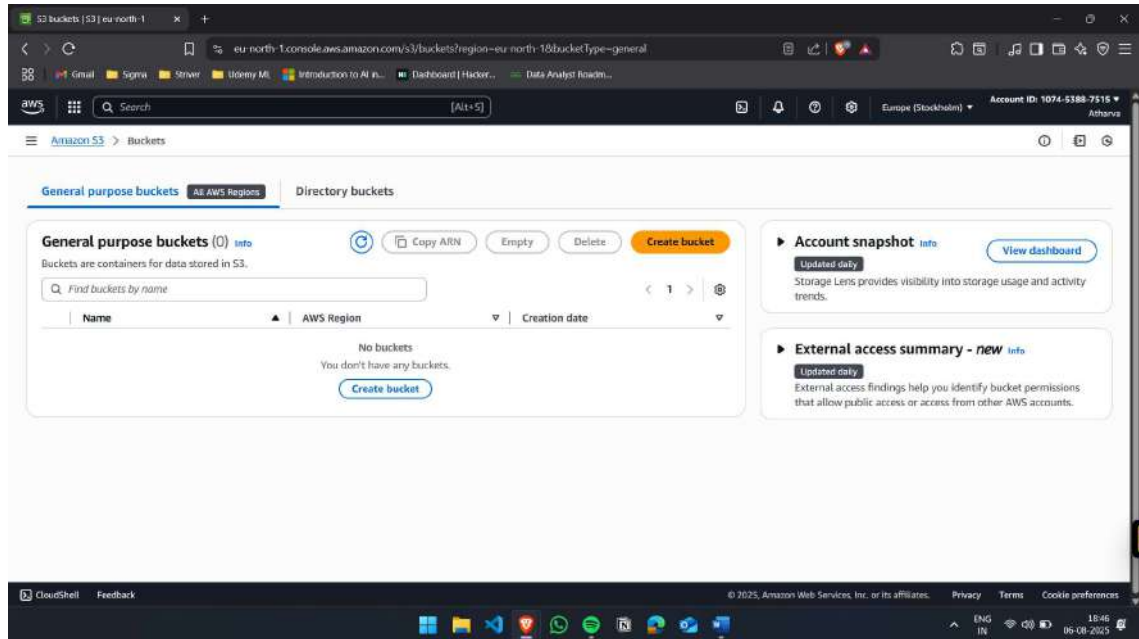
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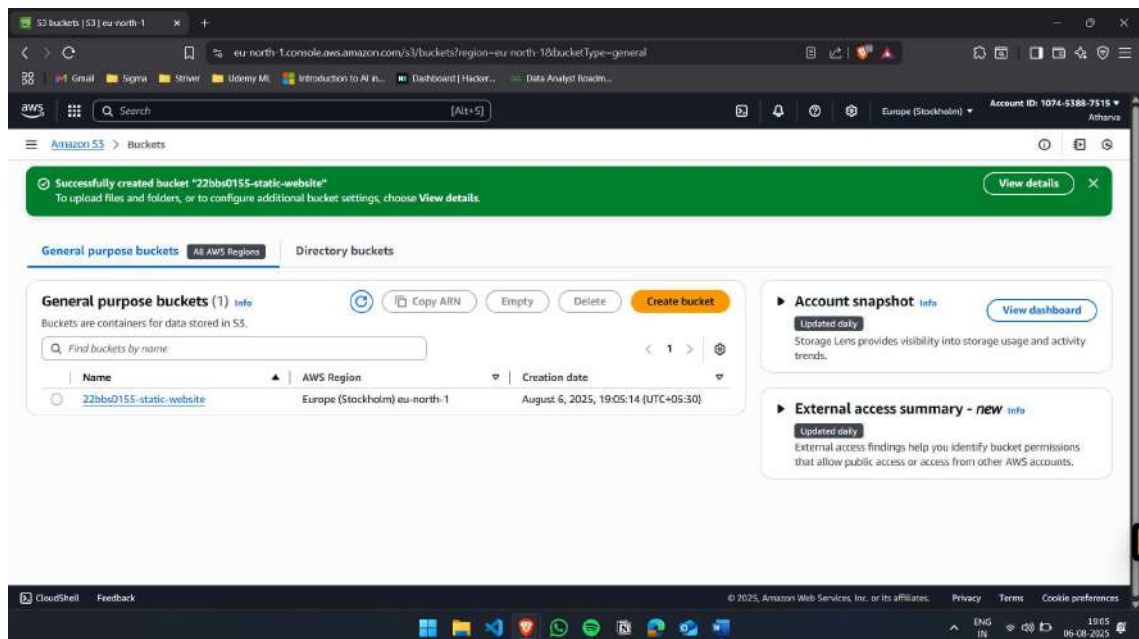
Q.2) Set up an S3 bucket in AWS, ensuring the name is globally unique. Upload your static website files (HTML, CSS, JavaScript, etc.) to this bucket. Enable static website hosting in the S3 bucket properties and configure the documents. Apply the necessary bucket policy to make your site publicly accessible. Also, enable versioning on the bucket to maintain previous versions of your website files.

Solution:

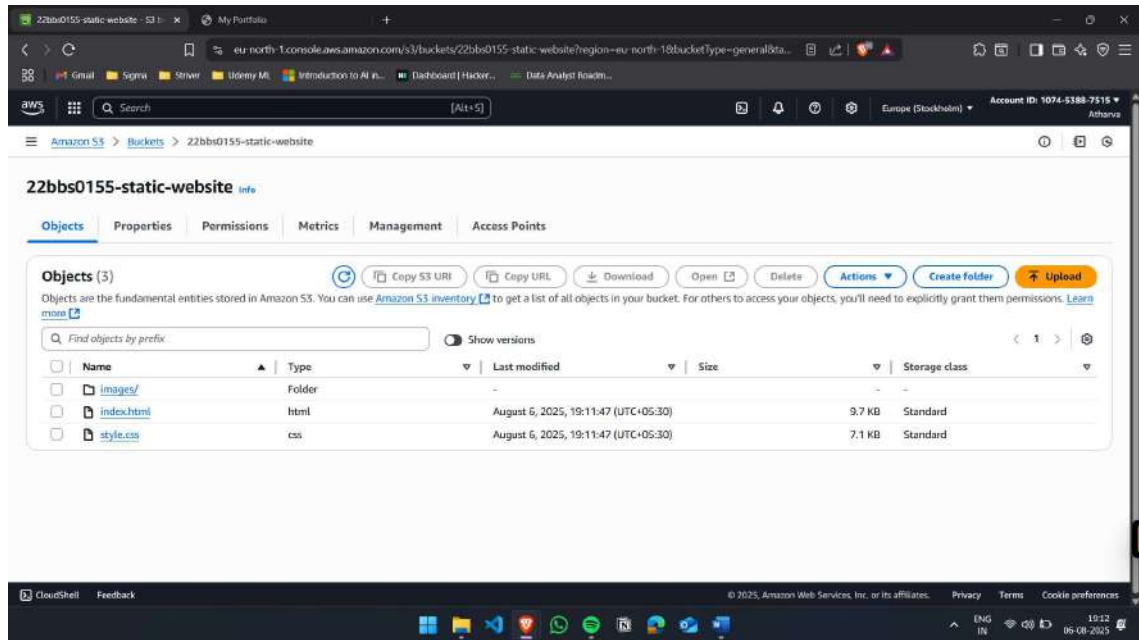
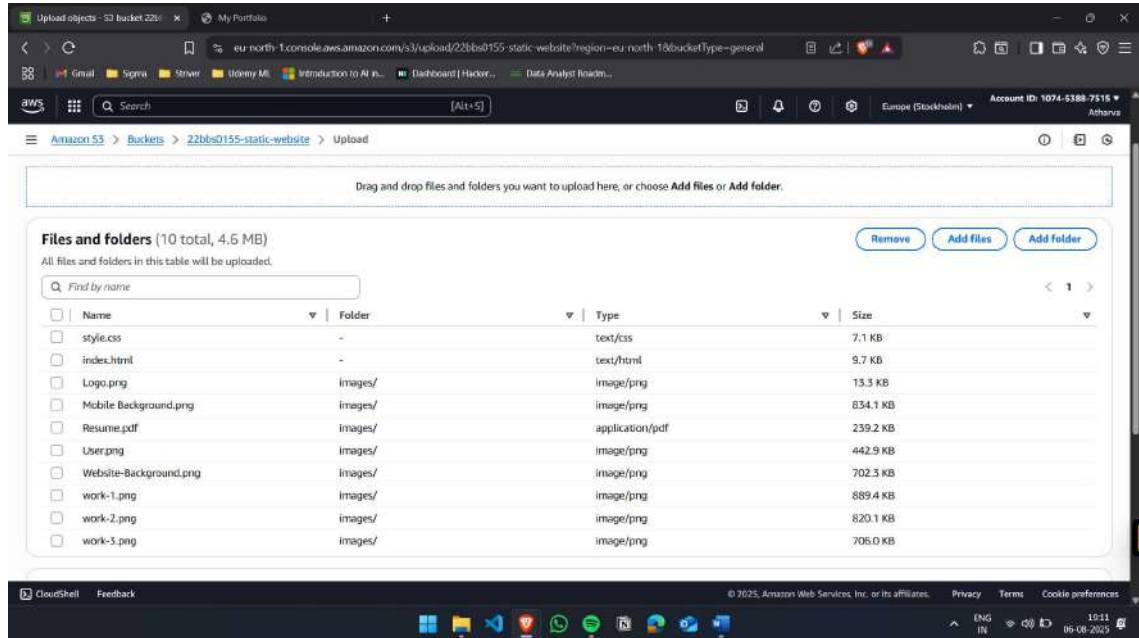
- Initial S3 Page:



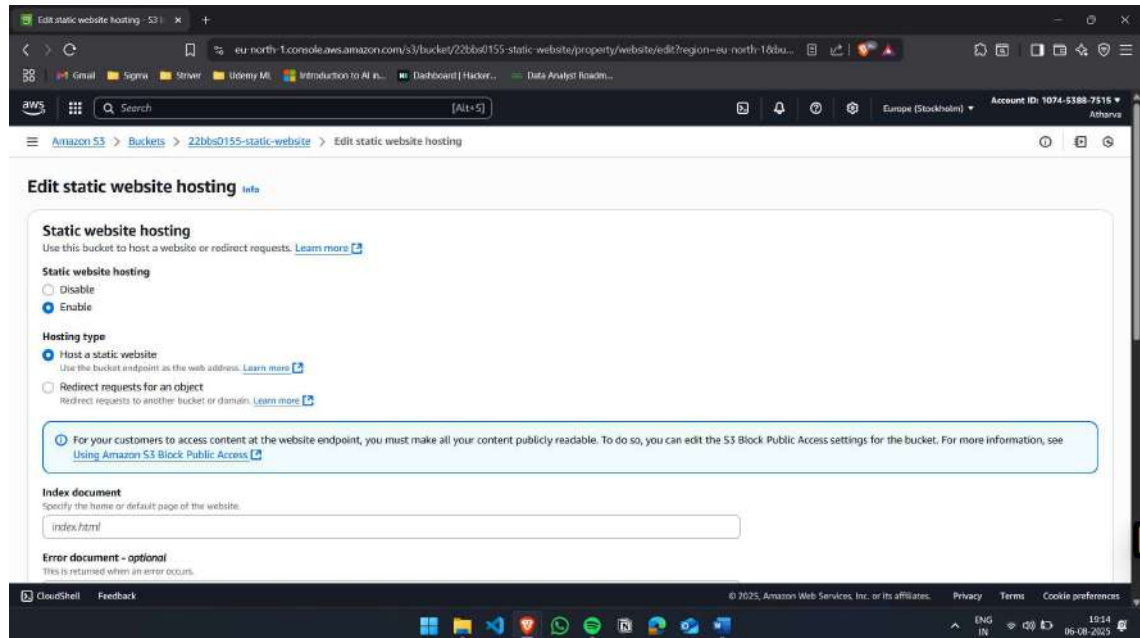
- Create S3 Bucket



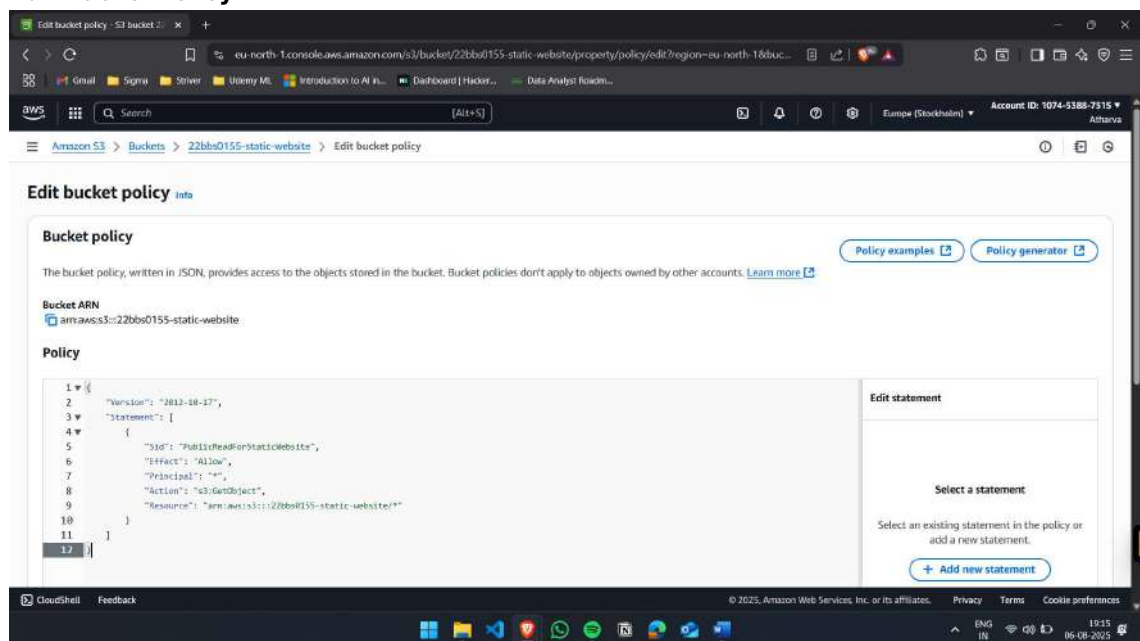
- Upload files



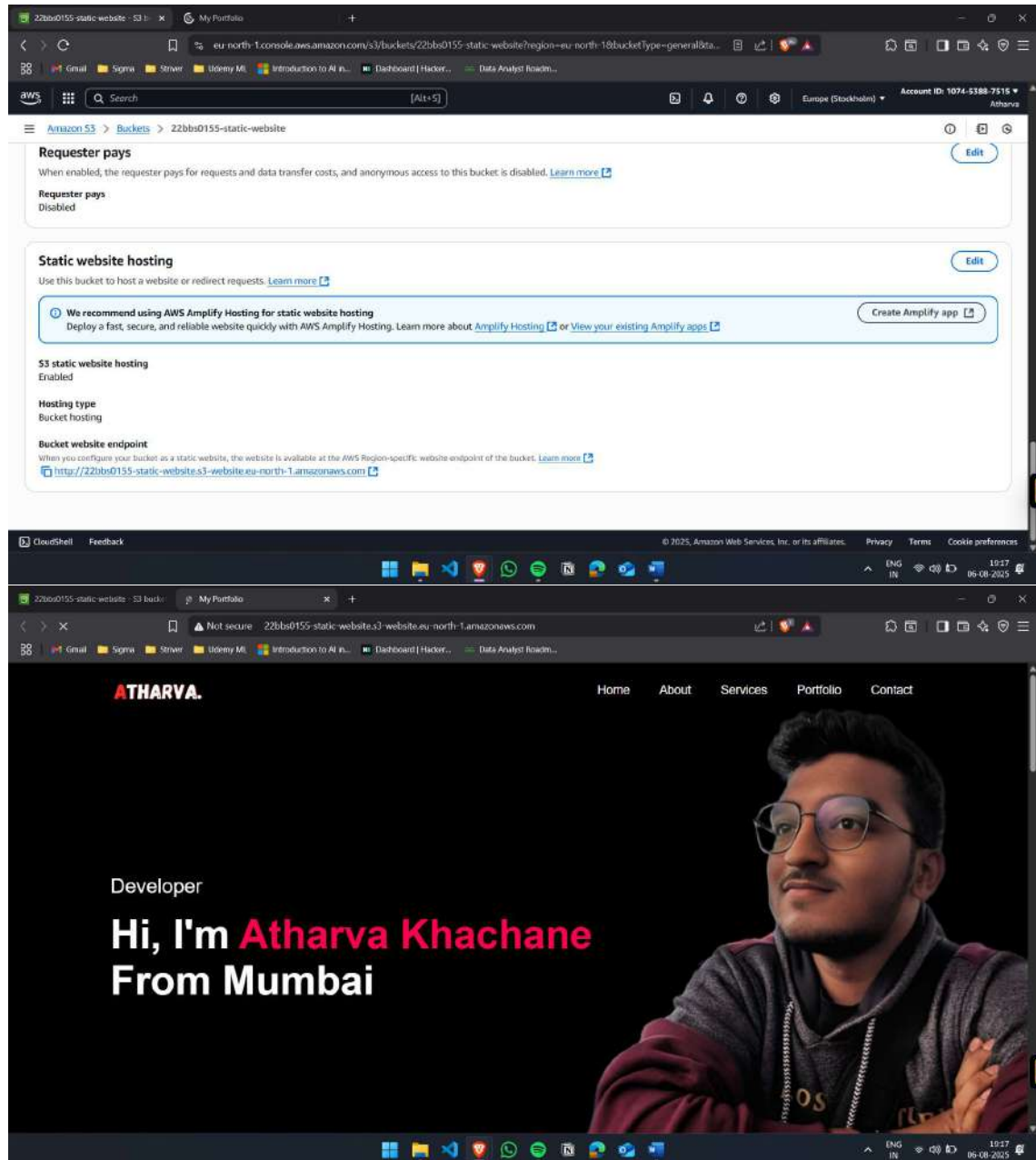
- **Enable Static Website Hosting**



- **Edit Bucket Policy:**



- Open the website



- **Enabled Versioning**

